

**Industry Classifications:
Distortions Caused by Systematizing**

September 2010

Following the meltdown in the financial markets in 2008, there has been an enormous effort across the financial services industry to improve methods of risk management. Widely employed tools of measuring risk, such as Value at Risk (VaR), have been called into question. For decades, the primary method of risk control in the investment management industry has been portfolio diversification. Indeed, central to modern portfolio theory and the efficient market theory is the idea that risk is most effectively managed through diversification across asset classes, industries, market capitalizations, and, in more recent years, geographies and currencies.

At Horizon, we believe proper diversification not only aids in managing risk within a portfolio, but also leads to superior returns over the long term. However, one must be certain that a portfolio is, in fact, diversified. This leads us to the subject of this commentary: distortions caused by broad classifications of securities. It is our view that the very practice of classifying securities, while useful in certain respects, ultimately leads to flawed conclusions regarding levels of portfolio diversification.

Imagine if you will, a scenario in which an investment strategist forecasts a flattening of the yield curve over the course of the next several quarters. The strategist therefore recommends to a portfolio manager that he underweight his exposure to stocks in the financial sector, as these businesses will naturally suffer from a decreasing net interest rate spread. The portfolio manager, trusting the instinct and analysis of the strategist, subsequently decreases his exposure to the following securities, all of which are classified, according to their GICS code, as financials: Brookfield Asset Management (BAM), NYSE Euronext (NYX), Henderson Land Development (12 HK), Leucadia National Corp. (LUK) and MasterCard (MA). Has this portfolio manager, having decreased his exposure to the financial sector, effectively adjusted his portfolio to benefit from a flattening of the yield curve?

Our short answer to this question is: no. Although he has reduced his portfolio weight in securities classified by GICS as financials, only a cursory review of the fundamental operations of the five aforementioned companies reveals that these are not, in fact, financials in the commonly understood sense of the term. A financial company, generally speaking, finances other parties, hence the term “financial”. Such a company is widely viewed as one that earns a return by borrowing at short-term rates and lending at higher long-term rates (and almost always by applying a substantial amount of leverage to greatly enhance that return). Consequently, the earnings and shareholders’ equity of these spread-based businesses are highly sensitive to changes in interest rates. However, this activity comprises only a minute percentage (if any percentage at all) of the operations of the five companies in question. Therefore, however brilliant the analysis of the investment strategist in the above scenario, the portfolio may not benefit because the classification of these companies is fundamentally flawed.

	<u>Debt-to-Equity Ratio</u>
Brookfield Asset Management	23%
MasterCard	0%
NYSE Euronext	35%
Henderson Land Development	23%
Leucadia National	50%
<i>Average for S&P 500 Financial Sector</i>	<i>330%</i>

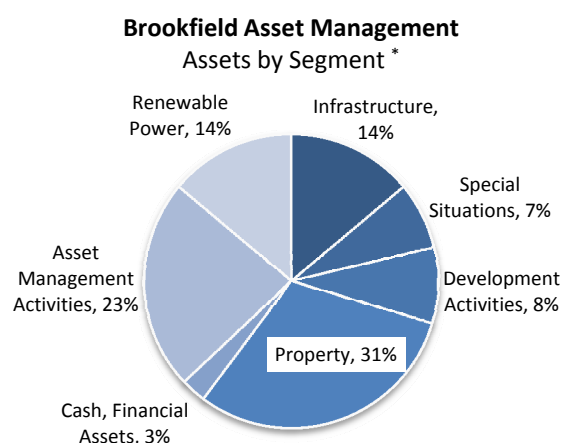
** Sourced from company reports and Bloomberg.*

The preceding example is one in which the portfolio manager willingly adjusts his own portfolio. However, it is often the case that these types of decisions are forced upon investment managers, who cannot stray from the sector allocations of a benchmark index. For example, a portfolio manager might find tremendous opportunity in the five aforementioned companies and buy them, but subsequently

receive a stern warning from her firm's risk control department that her allocation to the financial sector has grown too large. In this scenario, the portfolio manager initially increased the diversification of her portfolio, as these companies engage in a broad range of industries. However, by then selling these companies, the portfolio's diversification has been reduced in the name of risk management.

The essential problem with classification systems¹ is that they often misrepresent the actual operations of a company. These systems can mistakenly label a company as belonging to an industry, even if only a small percentage of its actual operations take place in that industry. Alternatively, they can misrepresent a company entirely, if it is more accurately represented by another industry. Those who are familiar with Horizon might have recognized that the five aforementioned companies are all substantial holdings across our investment strategies. Let us briefly review, then, the actual operations of these companies, and determine if they are accurately classified as financials.

Brookfield Asset Management is a global 'portfolio company' investing in high quality, long-lived assets. Roughly three-quarters of its assets are represented by 1) the largest hydroelectric power plant portfolio in North America; 2) Class A office buildings² in major international cities (e.g., New York, Washington, D.C., Sydney, and Toronto); 3) electric power transmission systems; and 4) timberlands. Furthermore, Brookfield is decidedly unleveraged, with a debt-to-capital ratio of only 15%, and the overwhelming majority of its debt (approximately 90%) is at the project level, on a long-term, investment grade, non-recourse basis. We ask somewhat rhetorically, then, if three-quarters of Brookfield's total assets are represented by renewable power operations, investments in office properties, and various infrastructure (utilities, timberlands) assets, and the company employs only a minimal amount of leverage, is it accurately classified as a financial company?



* Sourced from Brookfield Asset Management 2009 annual report.

MasterCard is understood by many to be a credit card lender, which it is not. While credit card companies such as Capital One Financial engage in lending (generating revenue from its net interest spread), MasterCard is simply a transaction processor, earning a small percentage of a credit card transaction's dollar value. The company's revenue is largely dependent upon the amount of retail sales activity, and therefore might more accurately be classified as belonging to the retail industry. Alternatively, MasterCard could easily be classified as a technology company; for example, another

¹ The GICS system was developed by MSCI and Standard & Poor's in 1999, and these two entities continue to assign companies to industries. The SIC system of classification was developed by the Central Statistical Board of the United States in the 1930s. Under the SIC system, no central agency assigns codes to individual businesses; instead, different entities assign SIC codes based on their own interpretation. Sourced from the Library of Congress: <http://loc.gov>.

² The occupancy rate of Brookfield's worldwide portfolio of office properties was 95% at June 30, 2010, 95% at year-end 2009, 97% at year-end 2008, 97% at year-end 2007, and 96% at year-end 2006. Despite the problems in real estate over the past three years (and, if one were to believe the countless media accounts, an impending crisis in commercial real estate), the effect upon the occupancy rates of Brookfield's office buildings has been negligible.

transaction processing company, Global Payments, is classified by GICS as belonging to the information technology industry.

NYSE Euronext may, at first glance, appear to be a classic financial company, as securities exchanges are often thought of as the quintessential capitalist enterprise. However, in point of fact, the main business of a securities exchange involves neither borrowing nor lending, nor taking any meaningful financial risk; rather, exchanges operate merely as facilitators of transactions. As the vast majority of exchange transactions are conducted electronically, an exchange might accurately be classified as a technology company. Furthermore, roughly 8% of NYSE Euronext revenue³ (and a significantly growing portion) is derived from its Market Data business, which involves the dissemination of real-time securities prices. This business is akin to the business of Bloomberg, or the publicly-traded Thomson Reuters, which itself is classified by GICS as belonging to the Consumer Discretionary sector (and in the Publishing sub-sector). Yet another growing portion of NYSE Euronext revenue is derived from the leasing of floor space to high-speed, high-frequency quantitative traders. These traders, wishing to have their computers located as close as physically possible to the data center of the exchange in order to gain a perhaps nanosecond advantage over competitors, have transformed the data center of the exchange from a cost to a profit center. Again, we ask rhetorically, is the NYSE Euronext, as a mere facilitator of transactions, a provider of market data, and a lessor of floor space, a true financial company in the sense of its operating earnings and balance sheet risks being tied to the short-term fluctuations in the cost of money or the debt-paying capacity of its customers?

Henderson Land Development is one of the largest real estate developers in Hong Kong. While real estate firms are often thought of as a subset of financials, their operations differ from the business of borrowing and lending, except the degree that the real estate they purchase is financed by debt. A sufficiently high degree of leverage transforms a real estate company into a “financial” in that such a business exposes itself to much the same risk variables as a true financial company. However, while it is not unusual for a real estate developer in the U.S. to have debt totaling 2x or 3x equity, at June 30th, the net debt to equity of Henderson Land Development was just 0.23x.⁴ More significantly, Henderson Land derives more than 40% of its net profit from its publicly owned subsidiaries, the largest of which is The Hong Kong China and Gas Company (3 HK). This company, founded in 1862 and commonly known as Towngas, is the sole provider of natural gas in Hong Kong. Towngas also has several joint ventures in gas supply on mainland China, where it has also begun to enter the businesses of supplying drinking water and waste disposal. In 2009, Towngas generated a net profit of HK\$5.3 billion, which amounted to almost 35% of Henderson Land’s net profit⁵.

The business of **Leucadia National Corp.**, much like Brookfield Asset Management, centers on the strategic allocation of capital. Its founders and current managers, Messrs. Ian Cumming and Joseph Steinberg, control over 20% of its shares, and managed to compound book value at 16.7% per annum over the 20 years from 1989 to 2009.⁶ Leucadia is comprised of a wide range of businesses, including manufacturing, telecommunications, land based contract oil and gas drilling, property management and services, gaming entertainment, medical product development and winery operations. The company

³ During the NYSE Euronext 2010 2nd quarter. Sourced from company reports.

⁴ Henderson Land Development 2010 Interim Results. http://www.hld.com/en/pdf/investor/circulars/2010/ann20100826_2_hld_e.pdf

⁵ Hong Kong China and Gas Company is approximately 40% owned by Henderson Land Development Co.

⁶ Sourced from Leucadia National 2009 annual report.

also has a significant investment in Jefferies Group, a full service investment bank, as well as very substantial equity interests in iron ore, gold and copper mining. The classification of Leucadia is particularly illuminating because Leucadia is often classified in entirely different industries. While GICS classifies Leucadia as a financial, the Securities and Exchange Commission assigns it a Lumber and Wood Products SIC code.⁷ The accompanying table breaks down Leucadia's sources of revenue for the last full fiscal year. The reader can ascertain whether GICS, SIC or, perhaps, neither is correct.

Leucadia National Corp.

2009 Sources of Revenue

Idaho Timber	12.8%
Conwed Plastics	7.3
Telecommunications	38.1
Oil & Gas Drilling Services	5.4
Property Mgmt & Services	10.5
Gaming Entertainment	9.3
Domestic Real Estate	2.7
Medical Products	0.5
Corporate & Other	13.5

It is important to note that this problem of misclassification is not limited to the categorization of industries; portfolios are often analyzed in terms of geographic exposure, with a company classified according to where its headquarters are located. Yet, oftentimes, where a company is domiciled has little or even no relation to where it conducts business. A prime example of this is illustrated by another Horizon holding, Phillip Morris International (PM), which is domiciled in New York and is therefore categorized by most portfolio analytical tools as a U.S. company. Yet, following its spin-off from Altria in 2007, Phillip Morris is *prohibited* from selling its products in the U.S. Would, then, a portfolio manager wishing to increase exposure to the United States and its economy achieve this through the purchase of New York based Phillip Morris?

In an attempt to simplify investing, emphasis is often placed on efficient categorizations or metrics. In the world of the physical sciences, one can classify items according to hard and fast rules: animals belong to a specific species, genus, order, class, phylum, and so forth. In the world of social sciences, the lines are often blurred and strict categorizations cannot be made. When attempts are made to categorize and systemize companies, significant distortions arise. This problem highlights the conclusion that Horizon has often discussed in past publications—that no social science can ever be understood in purely quantitative terms.

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⁷ Sourced from SEC website. <http://www.sec.gov>